TaskFlow Assistant (TFA) Module – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Module:** TaskFlow Assistant (TFA) · **Child Modules:** Chat, Actions and Tools, Help Knowledge, Audit Log and Rate Limits, Voice Input **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

> ✅ Target framework confirmed: `net10.0` (.NET 10), from `backend/src/TaskFlow.Api/TaskFlow.Api.csproj`.

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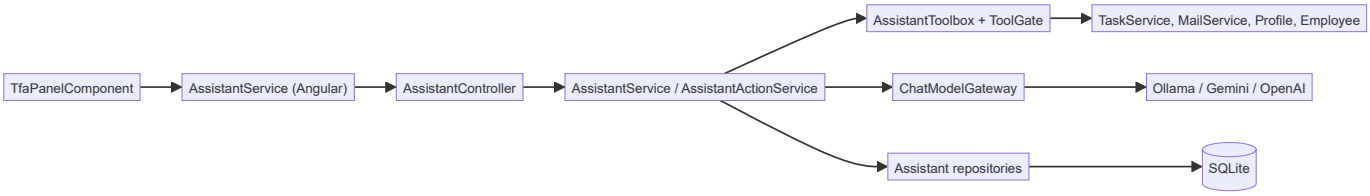
1. Technical Overview

TFA is a tool-calling chat assistant hosted inside the TaskFlow API. One controller (`AssistantController`, route prefix `api/v1/assistant`) exposes the signed-in user's own conversation, the actions TFA has prepared, and the list of AI providers the user may choose.

Main responsibilities:

- **Conversation engine** (`AssistantService`): validates a message (length, `SecretScreen`), applies the per-user rate limit, builds the system prompt (`AssistantPrompt`), then runs a bounded tool loop (max 4 tool rounds) against the selected provider through `IChatModelGateway`. Nothing is persisted until the provider has answered. → 09.1
- **Provider abstraction** (`Integrations/Ai`): `IChatModelProvider` implementations for Ollama (`/api/chat`), Google Gemini (`v1beta/models/{model}:generateContent`) and OpenAI (`v1/chat/completions`). Every network/HTTP problem is classified into a `ProviderFailureKind` and surfaced to the UI as a fixed message, never the raw provider body. API keys are encrypted with ASP.NET Core Data Protection (`AiKeyProtector`) and only decrypted in memory for one call. → 09.1
- **Tools** (`Services/Assistant/Tools`): 9 `IAssistantTool` implementations (6 read/guidance tools, 3 action-preparing tools). `AssistantToolbox` advertises only tools the caller may use and re-checks authorization on every call through `AssistantToolGate` → `IAssistantAuthorizer`, which evaluates the API's own `perm:{key}` authorization policies. → 09.2
- **Pending actions** (`AssistantActionService`): action tools only create an `AssistantPendingAction` (10-minute lifetime). Only the user's `POST actions/{id}/confirm` executes it, after re-checking the policy with the caller's current token and atomically claiming the row (`TryClaimAsync`), so it runs at most once. → 09.2
- **Help knowledge** (`HelpKnowledgeBase`): 19 embedded markdown topics with front-matter (title, roles, others, keywords, link), keyword-scored search, role-aware redaction. → 09.3
- **Audit and limits**: `AssistantAuditLog` writes `AssistantAudits` rows (no bodies/keys); `AssistantRateLimiter` is an in-memory sliding 1-minute window of 20 requests per user. → 09.4
- **Frontend**: a standalone Angular panel (`TfaPanelComponent`) mounted in the shell for users holding `assistant.chat`, with a signal-based `AssistantService`, action cards, model selector, cloud-provider notice and browser speech recognition / synthesis (wake phrase "Hey TF assistant"). → 09.1, 09.2, 09.5

2. Architecture Flow



3. Backend File Structure

Shared files (used by several children) are listed in full; child-specific files are listed in each child document.

Layer	Path	Files	Child
Controller	`TaskFlow.Api/Controllers/Assistant/`	`AssistantController.cs` (all endpoints; `RunActionAsync` helper for action endpoints)	09.1, 09.2
Service – Caller	`TaskFlow.Api/Services/Assistant/Access/`	`AssistantCaller.cs` (also `AssistantPolicies`)	Shared
DTOs	`TaskFlow.Api/Dtos/Assistant/`	`AssistantDtos.cs`, `AssistantActionDtos.cs`, `AssistantFailureDto.cs`	09.1, 09.2
Entities	`TaskFlow.Api/Models/Entities/`	`AssistantConversation.cs`, `AssistantMessage.cs`, `AssistantPendingAction.cs`, `AssistantAudit.cs`	09.1, 09.2, 09.4
Enums	`TaskFlow.Api/Models/Enums/`	`AssistantActionKind.cs`, `AssistantActionStatus.cs`, `AssistantMessageRole.cs`, `AiProviderKind.cs`, `ProviderFailureKind.cs`	09.1, 09.2, 09.4
Authorization	`TaskFlow.Api/Authorization/Assistant/`	`IAssistantAuthorizer.cs`, `AssistantAuthorizer.cs`	09.2
Exceptions	`TaskFlow.Api/Common/Exceptions/`	`AssistantExceptions.cs` (`AssistantRequestException` → 09.1, `AssistantRateLimitedException` → 09.4, `AssistantActionConflictException` → 09.2)	Shared
DI	`TaskFlow.Api/Config/`	`ServiceRegistration.cs` (Assistant block), `RepositoryRegistration.cs` (Assistant block), `IntegrationRegistration.cs` (Data Protection, provider `HttpClient`s), `AiProviderOptions.cs`	Shared
Child-specific	`Services/Assistant/Chat/`, `ModelGateway/`, `Integrations/Ai/`, `Repositories/Assistant/Conversations/`, `Messages/`	see `09.1-Chat.md`	09.1
Child-specific	`Services/Assistant/Actions/`, `Tools/`, `Toolbox/`, `Repositories/Assistant/PendingActions/`, `People/`	see `09.2-Actions-and-Tools.md`	09.2

Layer	Path	Files	Child
Child-specific	`Services/Assistant/HelpKnowledge/`, `Knowledge/`	see `09.3-Help-Knowledge.md`	09.3
Child-specific	`Services/Assistant/AuditLog/`, `RateLimits/`, `Repositories/Assistant/Audits/`	see `09.4-Audit-Log-and-Rate-Limits.md`	09.4
Rules	`TaskFlow.Api/Services/Assistant/Rules/`	`AssistantRules.cs`, `SecretScreen.cs` → 09.1; `AssistantActionRules.cs`, `TaskPrioritiser.cs` → 09.2	09.1, 09.2

4. Frontend File Structure

Type	Path	Purpose	Child
Panel component	`src/app/modules/assistant/components/tfa-panel.component.ts/.html`	Floating TFA button, panel shell, cloud notice, feed, failure card, voice wiring	09.1, 09.5
Child components	`tfa-header`, `tfa-composer`, `tfa-message` (`components/`)	Header, composer, message rendering	09.1 (composer mic → 09.5)
Child components	`tfa-action-card` (`components/`)	Pending action card	09.2
API service	`src/app/modules/assistant/services/assistant.service.ts`	All HTTP calls to `/api/v1/assistant`	09.1, 09.2
Voice services	`services/wake-phrase.service.ts`, `speak.service.ts`	Speech recognition / synthesis	09.5
Models	`src/app/modules/assistant/models/assistant.model.ts`, `services/assistant.model.ts`, `services/wake-phrase.model.ts`, `services/speak.model.ts`	TypeScript interfaces matching the DTOs	Shared
Mount / guard	`src/app/layouts/shell.component.ts/.html`	`@if (canUseAssistant())` renders `app-tfa-panel`; `canUseAssistant` = user permissions include `Permissions.AssistantChat`. Shell route is behind `authGuard`, `mfaSetupGuard` (`routes/app.routes.ts`)	Shared
Permission constants	`src/app/shared/auth/permissions.ts`	`AssistantChat: 'assistant.chat'`, `AssistantActions: 'assistant.actions.confirm'`	Shared

There is no TFA route and no menu entry in `layouts/nav-items.ts` (only the separate **TFA Settings** entry, which belongs to the Settings module).

5. Child Modules

#	Child Module	Main endpoints (count)	Key permissions	Document
09.1	Chat	`GET conversation`, `POST conversation/messages`, `POST conversation/new`, `GET providers` (4)	`assistant.chat`	`09.1-Chat.md`
09.2	Actions and Tools	`PUT actions/{id}`, `POST actions/{id}/confirm`, `POST actions/{id}/cancel` (3) + 9 model tools	`assistant.actions.confirm`, `roles.manage`, `assistant.chat`	`09.2-Actions-and-Tools.md`
09.3	Help Knowledge	none – `search_help` tool inside `POST conversation/messages` (0)	`assistant.chat`	`09.3-Help-Knowledge.md`
09.4	Audit Log and Rate Limits	none – guards `POST conversation/messages`, writes audit rows (0)	–	`09.4-Audit-Log-and-Rate-Limits.md`
09.5	Voice Input	none – frontend only, uses `POST conversation/messages` (0)	`assistant.chat`	`09.5-Voice-Input.md`

All endpoints are under `/api/v1/assistant` and carry the class-level `[Authorize]` (JWT). Child documents are in the `09-Assistant/` folder next to this file.

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6. Database Design

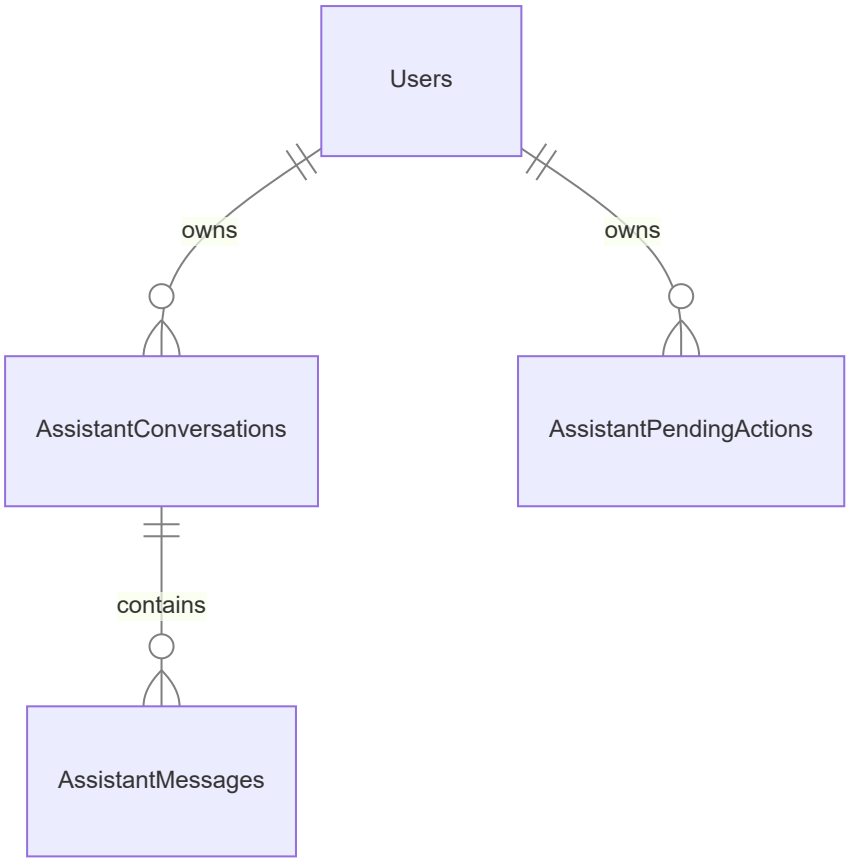
Four TFA tables, created by migration `Persistence/Migrations/20261018183210_AddAssistant.cs` (which also creates `AiProviderConfigs`, owned by the Settings module). Configuration in `Persistence/AppDbContext.cs` (`modelBuilder.Entity<AssistantConversation>` … `AssistantAudit`).

Table	Purpose	Owned by child
`AssistantConversations`	One row per conversation; latest by Id is the active one	09.1 Chat
`AssistantMessages`	User and assistant messages with `MetaJson` (kind, links, ranked tasks, action id)	09.1 Chat
`AssistantPendingActions`	Prepared actions and their status	09.2 Actions and Tools
`AssistantAudits`	Action outcome audit rows	09.4 Audit Log and Rate Limits

Column-level detail is in each owning child document. 09.3 and 09.5 use no tables.

Relationships

- `AssistantConversations.UserId → Users.Id` (FK, Cascade)
- `AssistantMessages.ConversationId → AssistantConversations.Id` (FK, Cascade)
- `AssistantPendingActions.UserId → Users.Id` (FK, Cascade)
- `AssistantPendingActions.ConversationId` and `AssistantAudits.UserId` are plain columns without FK constraints.



7. Security & Authorization

- **Authentication:** JWT bearer token + refresh-token cookie; `[Authorize]` on `AssistantController`. `AssistantCaller` reads user id and role from the validated token claims only.
- **Authorization:** Permission policies `perm:{key}` registered for every `PermissionCatalog.All` entry (`Config/CrossCuttingRegistration.cs`, `RequireClaim(AuthClaims.Permission, key)`); keys in `Common/Security/PermissionKeys.cs` (`AssistantChat = "assistant.chat"`, `AssistantActions = "assistant.actions.confirm"`). TFA tools and confirmations are checked with the same policies through `IAssistantAuthorizer` (`IAuthorizationService.AuthorizeAsync`).
- **Defaults** (`PermissionCatalog.cs`): both TFA keys `GrantedToManager: true, GrantedToEmployee: true, GrantToAllExistingRoles: true`; Administrator holds all assignable entries.
- **Data scope:** Own – conversations and pending actions are always filtered by `UserId` (another user's id yields 404). Task data is read through `ITaskService.ListAsync / GetDetailsAsync(…, userId, role)`, so the Task Management scope (own / all tasks) applies.
- **Model containment:** the model can only call advertised tools; action tools only create pending actions; tool output is marked as data and passed through `SecretScreen.Redact`; refusals and confirmations use fixed text (`AssistantTexts`).
- **Secrets:** `SecretScreen` blocks passwords/keys in user text; API keys are encrypted with Data Protection purpose `TaskFlow.AiProviderKeys` (DPAPI-protected key ring on Windows); provider `HttpClient`s have logging removed and `AllowAutoRedirect = false`; Gemini key sent in `x-goog-api-key` header, OpenAI as Bearer header.
- **Frontend:** the panel is rendered only when the user has `assistant.chat` (shell). Action card buttons are not gated by `assistant.actions.confirm` (see 09.2 Gaps).

Permission Key	Endpoint(s)	UI Element Controlled	Child
`assistant.chat`	4 conversation/provider endpoints	Floating TFA button and whole panel (`shell.component.html` `@if (canUseAssistant())`)	09.1 (also 09.3, 09.5)
`assistant.actions.confirm`	3 action endpoints	None in UI	09.2
`roles.manage`	Tool `propose_role_change`; re-checked in `ConfirmAsync`	Tool advertised to the model only when held	09.2

8. Validations & Error Handling

Shared rules only; child-specific rules are in each child document.

Rule	Layer	HTTP Code	Error Message
No permission (endpoint)	Policy	403	Forbidden
Unhandled error	`GlobalExceptionHandler`	500	Something went wrong. Please try again.

Errors are returned by `Middleware/GlobalExceptionHandler.cs` as `application/problem+json`: `AppException` → its `StatusCode`, `Title` (`TFA could not handle that`, `Too many requests`, `This action can no longer be done`) and message; `RetryAfter` sets the `Retry-After` header. The Angular service shows the problem `detail` via `apiErrorMessage` with per-call fallback texts (listed in 09.1 and 09.2).

9. Integrations / Side Effects

- **AI Assistant:** this is the AI Assistant module; the tools it exposes are listed in `09.2-Actions-and-Tools.md` §5.1.
- **Notifications / Email / Profile / Employees:** through confirmed actions – see 09.2.
- **AI providers and key-ring startup check:** see 09.1.
- **Activity Log:** TFA's own audit rows – see 09.4. Endpoint-level 403s on any TFA endpoint are also recorded by the application's `DeniedActionAuditHandler`.

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Application:Timezone`	Fallback time zone for "today" / overdue when no application setting is stored (`AppClock`); used by the system prompt (09.1) and task ranking (09.2)	`Asia/Kolkata`
`ConnectionStrings:Default`	SQLite database	`<secret>`

Child-specific keys and code constants: `AiProviders:*`, `Storage:Root`, `AssistantRules.*`, frontend provider/cloud keys → 09.1; `AssistantActionRules.Lifetime` → 09.2; `HelpKnowledgeBase.MinimumScore` → 09.3; `AssistantRateLimiter.RequestsPerMinute` → 09.4; `tfa.handsfree` → 09.5.

11. Testing

Test Type	File / Location	Covers
Integration	`tests/TaskFlow.IntegrationTests/Assistant/AssistantEndpointsTests.cs`	Assistant endpoints + authorization (all children)

Child-specific test files are listed in each child document. (Coverage descriptions are inferred from file names only.) Frontend specs: `tfa-panel.component.spec.ts`, `assistant.service.spec.ts`, `speak.service.spec.ts`, `wake-phrase.service.spec.ts`.

12. Known Limitations / Future Enhancements

Child-specific gaps are listed in each child document.

Module-wide gaps found

- **Requirements and design coverage:** TFA is not one of the 20 original screen specs. It is covered by the approved BRD `TF - Documents/TF - BRD/08-TFA-Assistant/08.1-Assistant/BRD-Assistant.md` and by TDD §16 (TFA Assistant Design).
- **Project file:** `TaskFlow.Api.csproj` targets `net10.0` and embeds `Services\Assistant\Knowledge\*.md` (verified).

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